

Individual differences, immunity, and cancer: Lessons from personality psychology

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Abstract

Individual differences have an important role in modulating the relationship between environmental events such as stressors and the immune system, and stable individual differences such as personality may have an important influence not only on immunity but on the progression of chronic health problems such as cancer. However, the relationships among personality, the immune system, and health—especially cancer—are controversial. Looking back to another controversy may help inform future research in this area: In the 1960s, Walter Mischel questioned whether stable personality traits existed. Responses to this challenge included research designs that improved reliability and accounted for both person and situation in an interactionist model. Both responses are relevant to current research in personality, immunity, and health.

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1. Introduction

Hundreds of published studies have described relationships between psychosocial stressors and the immune system. However, immunocytes are themselves insensible to psychosocial events; only the central nervous system can transduce psychosocial stressors into signals that can influence the immune system. Stressors are characterized as such and often treated as monotonic because the majority of individuals perceive them as stressful. However, there are important differences in the ways people perceive and respond to environmental events and even internally generated events (e.g., imagining giving a speech in front of hundreds of people). Individual differences in cognition, affect, and behavior, therefore, have a potentially important role in modulating the immune system.

The relevance of these relationships to the onset or progression of cancer, however, is tenuous. It is possible

that personality can affect cancer via the immune system. This hypothesis has not received much empirical support, but studies may have been obscured by methodological weaknesses. The purpose of this review is to suggest some methodological approaches from personality psychology that could lead to stronger tests of this hypothesis.

2. The cancer controversy

Individual differences and immunity have both been implicated in the initiation and progression of cancer (see, e.g., Butow et al., 2000; Garssen & Goodkin, 1999; Spiegel & Kato, 1996; Souberbielle & Dalglish, 1994; Turner-Cobb, Sephton, & Spiegel, 2001, for reviews). In the area of individual differences, it has been proposed that personality characteristics reflecting emotional suppression, helplessness, fatalism, and a “Type C” constellation including stoicism, perfectionism, and over-agreeableness are risk factors for the initiation and

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progression of cancer (e.g., Greer, 1991; Gross, 1989; Temoshok, 1987). While research tends to support such a relationship, the evidence is not incontrovertible, and some investigators have countered that psychosocial effects are negligible (e.g., Cassileth, Lusk, Miller, Brown, & Miller, 1985; Jamison, Burish, & Wallston, 1987).

Even more controversial is the role of the immune system as a mediator between personality and cancer outcomes. In this model, personality affects immune function, particularly in response to stressful life circumstances, and those changes in immune function lead to greater vulnerability to cancer initiation or progression. There is evidence that personality is associated with immune parameters both on an ongoing basis and in response to stressors (Segerstrom, 2000). However, those immune differences may or may not be of the type and magnitude to affect cancer outcomes; furthermore, the anti-tumor potency of the immune system is modest (Garssen & Goodkin, 1999; Turner-Cobb et al., 2001). Anti-tumor activity by the immune system can be limited by poor immunogenicity of the tumor, anti-immune activity of the tumor, and lack of costimulation by the tumor. However, there is evidence that the immune system can be active in fighting cancer: Immunotherapy is successful in treating some types of cancers, including melanoma and B cell malignancies, and natural killer cells may play a role in immunosurveillance and cytotoxicity against metastatic disease (Armstrong, Eaton, & Ewing, 2001; Pardoll, 2001; Whiteside & Herberman, 1995).

3. Methodology: A source of controversy

Adding to the confusion and controversy in this literature is methodological heterogeneity. Diverse samples, methods, and models have not yielded converging evidence for the relationships among personality, immunity, and cancer outcomes (Temoshok & Heller, 1984). For example, many studies have used mixed cancer samples, though different types and stages of cancer differ biologically in important ways and therefore may be differentially affected by psychosocial and immune factors. As another example, many studies lack a true prospective design, leading to difficulties in causal interpretations.

A third methodological challenge has to do with the time frame of cancer. Unlike many infectious diseases, whose onset and clinical course can be measured in hours or days, the time frame of cancer is best measured in longer time frames. A clinically detectable cancer has likely been growing for a period of years (in fast-growing cases such as acute leukemias) or decades (in slow-growing cases such as breast, prostate, or colon; Friberg & Mattson, 1997). Personality is among the most

promising predictors of cancer outcomes because it can exert an influence on the immune system over the long periods of time that correspond to the growth period of cancer. However, there are methodological advances in personality research that have yet to be applied to the relationships among personality, immunity, and cancer.

4. A second controversy: Does personality exist?

These methodological advances arise from an older controversy that involved the existence of personality *per se*. While people intuitively believe that they act consistently and therefore have personality, this premise has been challenged. Walter Mischel, a social psychologist, noted in the late 1960s that personality accounted for less than 10% of the variance in behavior (Mischel, 1968, 1969). He further suggested that the perception of continuity in behavior was a “cognitive construction... only very tenuously related to the phenomena that are construed” (Mischel, 1969, p. 1012); in essence, Mischel asserted that personality did not exist. These assertions naturally caused something of a crisis in personality research and theory. More importantly for the purposes of this paper, they provoked a series of responses that are relevant to research on personality, immunity, and cancer. These responses included theoretical, methodological, and statistical viewpoints focusing on reliability (Epstein, 1979) and interactionism (Bowers, 1973).

5. The reliability approach

The reliability approach to consistency in personality assumes that there are multiple inputs to affect, cognition, or behavior at any given time, some of which reflect personality and some of which do not. For example, how I feel at this moment is not only a function of my stable affective predispositions, but also the temperature of my office, the interactions with my students over the afternoon, what I ate for lunch and with whom, how well I slept last night, and a myriad of other influences. Not surprisingly, then, personality is only modestly related to momentary mood. For example, in one study, trait anxiety accounted for only about 6% of same-day positive or negative affect (Segerstrom, 2001a). Likewise, there is evidence that affect varies widely from day to day, with only 25–35% overlap (Epstein, 1979).

Are there, then, affective dispositions? Are there chronically happy and sad people? Personality constructs with affect at their core, such as neuroticism, extraversion, and negative affectivity, are among the most basic constructs of trait personality theories (John & Srivastava, 1999; Watson & Clark, 1984). One possibility is that such dispositions are, as Mischel argued, constructions rather than realities. Another possibility is

that single-point assessments are inadequate to reveal dispositions, and this is the crux of the reliability approach to personality. Over a large number of observations, transient influences on my mood (the temperature of my office, what I ate for lunch) will tend to diminish in magnitude because the variations cancel each other out. On the other hand, stable characteristics such as personality—those that are consistently present at each assessment—will have the same influence on each day. As more days are considered, these stable traits become more influential than momentary influences. Epstein (1979) demonstrated that, for positive and negative mood, reliability approached acceptable levels (generally .80) when data were aggregated over a number of days: 16 for positive mood and 24 for negative mood. He obtained similar findings for impulses (e.g., aggression, withdrawal), behavior (e.g., nurturance, physical escape), and, to a lesser degree, situations (e.g., isolation, inconsideration).

Immune parameters, likewise, may be influenced by transient psychosocial factors such as stressors or emotions (Futterman, Kemeny, Shapiro, & Fahey, 1994; Herbert & Cohen, 1993) as well as by stable personality factors such as extraversion or a repressive coping style (Jamner, Schwartz, & Leigh, 1988; Miller, Cohen, Rabin, Skoner, & Doyle, 1999). When one investigates the relationship of personality to immunity or immunity to chronic disease, it is often the stable variance in immune parameters that is of interest rather than the transient variance. The temporal reliability (i.e., stable variance) of immune parameters varies across measures: Some parameters, such as salivary IgA secretion and natural killer cell subsets in peripheral blood, change rapidly in response to short-term influences, including stressors (Kiecolt-Glaser, Cacioppo, Malarkey, & Glaser, 1992) and daily mood (Stone, Cox, Valdimarsdottir, Jandorf, & Neale, 1987). The reliability approach predicts that stable variance in measures such as these will be evident only when aggregated over a number of occasions, and this prediction is borne out for IgA. Stone et al. (1987) measured salivary IgA production to rabbit albumin 25 times over 8 weeks. The stable variance was only evident when several occasions were aggregated: Reliability increased from .20 when only two occasions were considered to a maximum of .70 when all occasions were considered (Stone, 1992).

Even relatively stable immune assessments can increase in reliability when multiple occasions are considered. Delayed-type hypersensitivity (DTH) skin test responses unfold slowly over a period of hours to days and therefore already represent an aggregate of psychological influences during that period. However, when DTH responses were taken on 8 occasions separated by 2 months, reliability increased from a low of .77 to a high of .90 (Lesourd, Winters, & Darnell, 1986). One concludes from these data that there are stable individ-

ual differences in immune parameters that may mirror personality characteristics and that may also affect long-term health; however, they are not always apparent from a single assessment.

Reliability limits the correlation between two measures: any two measures cannot correlate more highly than their reliabilities. Consider therefore the following model:

Personality → Immune parameter(s)
→ Cancer outcome(s)

Each of the arrows represents the relationship between that parameter and its neighbor. Personality is generally assessed very reliably (.80 or greater). Cancer outcomes such as recurrence and mortality are generally assessed very reliably. However, immune parameters are often not very reliable, and this will severely limit the magnitude of both of the relationships represented by the arrows above. To demonstrate the effects of increasing immunological reliability, consider an imaginary case in which a personality measure accounted for about 10% of the IgA variance on a single day. The low reliability in IgA (.20) may have masked the true influence of personality in this case. Correcting statistically for this unreliability yields a “true” relation that is substantially higher, accounting for about 45% of the IgA variance.

This approach is not only relevant to studies of personality but extends to any situation in which the stable variance in immune parameters is of interest. Psychological interventions are often predicated on the idea that their influence will last beyond the period of the intervention itself. For example, in psychoeducational interventions, participants are often expected to continue applying the knowledge and skills acquired in the intervention indefinitely. For example, Fawzy et al. (1993, 1990) administered a 6-week group intervention to melanoma patients in a randomized clinical trial; the intervention resulted in higher natural killer cell subsets and cytotoxicity (NKCC) at 6 months post-intervention and lower rates of recurrence and mortality at 5 years post-intervention. However, only baseline NKCC was significantly associated with recurrence, and neither baseline nor change in NKCC was associated with survival.

NKCC is a relatively volatile parameter. In a sample of depressed patients, NKCC assays showed only 25% stable variance over a 1-week interval, compared with 53% for enumerative measures and 42–53% for proliferative measures (Miller, Cohen, & Herbert, 1999). Fawzy et al. (1990) assessed NKCC on one occasion at each critical point. How well did this assessment approach represent stable changes in this parameter that could influence disease course over a matter of months or years? Statistical correction indicates that as NKCC reliability increases from .25 to 1.0, the size of the

obtained relationship between NKCC and another reliably assessed variable (e.g., cancer outcomes) increases to double its original magnitude.

It is, of course, more difficult and expensive to do multiple assessments of immune parameters to achieve adequate reliability. However, the potential reward is great. Research that invests in recruiting clinical samples and following them over long periods of time should not be defeated from the start by unreliability.

6. The interactionist approach

Applying the reliability approach to the relationship between personality, immune parameters, and disease will be productive when studying a stable, progressive process. However, there may be periods of time that are critical to disease outcomes. For example, surgery to remove metastatic tumors has been proposed as a critical period in which the metastatic process can be disinhibited or even facilitated, and the immunological response during this process may have as much or more influence on long-term outcomes than tonic immune status (Ben-Eliyahu & Shakhar, 2001). In this case, it will be most important to understand how personality affects immunity during this specific, critical situation rather than aggregating across a number of situations. The interaction between personality and situational variables is the basis of the interactionist approach. Bowers (1973) characterized the interactionist approach best when he wrote that “different situations affect different individuals differently” (p. 324). Bowers’s analysis of 11 studies testing the effects of person, situation, and their interaction supported Mischel’s assertion that person variance was small: 13% across all studies. However, situation variance was even lower at 10%. The person by situation interaction accounted for by far the most variance at 21%.

One personality characteristic that has been investigated in PNI and psycho-oncology is optimism, operationalized as positive outcome expectancies, and this literature offers the opportunity to demonstrate the power of interactionism. Optimism is generally considered a protective personality trait that contributes to resilience under stress (Scheier & Carver, 1985). Consistent with this conceptualization, optimism has been associated with higher immune parameters during stressors (Byrnes et al., 1998; Cohen et al., 1999; Segerstrom, Taylor, Kemeny, & Fahey, 1998, in press; Sieber et al., 1992). These results suggest that optimism buffers individuals against negative immunological effects of stressors. However, most of the studies reviewed above also found negative effects of optimism (Cohen et al., 1999; Segerstrom, 2001a; Segerstrom et al., in press; Sieber et al., 1992). These different effects arose from the specific conditions under which optimism was

studied. When the stressor was straightforward or responsive to coping efforts, optimism was associated with higher immune parameters; when conditions were difficult or nonresponsive, optimism was associated with lower immune parameters. These differences are likely to be due to optimists’ greater propensity to stay engaged with stressors in the face of nonprogress (Carver & Scheier, 1998). In the short term, engagement with stressors can be physiologically more stressful than giving up (Mullen & Suls, 1982; Suls & Fletcher, 1985). Optimists are willing to tolerate this stress because they expect an eventual positive outcome, whereas pessimists are less likely to tolerate it, not anticipating an eventual reward. The interaction between optimism and situations is critical to the nature of the relationship between optimism and immune parameters. Interactionism addresses the effects of personality on immune parameters in specific situations and is relevant to chronic disease to the degree that the disease is punctuated by critical periods for progression (e.g., surgery for metastatic disease).

A second type of interactionism that may be relevant to chronic disease refers to the effect of personality on the kind of situations to which one is exposed. For example, over periods ranging from days to years, the personality trait of neuroticism (characterized by predisposition to negative moods, emotional instability, and behavioral impulsivity) is associated with greater incidence of negative interpersonal interactions, daily hassles, and life events such as divorce and financial loss (Bolger & Schilling, 1991; Headey & Wearing, 1989; Saudino, Pedersen, Lichtenstein, McClearn, & Plomin, 1997). Therefore, neuroticism predicts both greater emotional reaction to negative events and higher incidence of those events over time, both of which could affect the long-term progression of chronic disease.

7. Conclusion and future directions

The state of the literature in personality, immunity, and cancer progression is reminiscent of the state of personality psychology in the 1960s. Small and unstable relationships between personality and behavior led Mischel (1968, 1969) to question whether personality exists: Whether there is consistency in behavior that is predictive of future behavior. Similarly, there are small and inconsistent relationships between personality, immunity, and cancer that have led some to question whether these relationships are important. Garssen and Goodkin (1999) wrote, “Enthusiasm for the hypothesis that psychological factors might have some influence on the development and progression of cancer has waxed and waned. The present position is probably best described as stagnation; there is a need to think about the design of future studies to test this hypothesis more effectively,

especially since many studies have yielded negative findings.” (p. 51). Researchers in psycho-oncology may wish to draw on the lessons learned from the crisis in personality psychology to advance knowledge of the effects of personality on immunity and health.

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